1) $2 \cdot 3 - 1 = 5$

2) $1.5 \cdot 1 - 0.5 = 1$ $\quad \frac{1}{1+e^{-1}} = 0.731$

3) $(3) - 2 + 1 = -6 + 1 = -5$ $\quad$ Relu $= 0$

4) $\frac{1}{2}(\hat{y} - y)^2 = \frac{1}{2}(3)^2 = \frac{9}{2}$

5) $\frac{\partial L}{\partial \hat{y}} = \hat{y} - y = 6 - 10 = -4$

6) $\hat{y} = wx + b = (1)(2) + 0 = 2$

$\frac{\partial L}{\partial \hat{y}} = 2 - 5 = -3$

$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial w} = -3 \cdot 2 = -6$

$\frac{\partial L}{\partial b} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial b} = -3 \cdot 1 = -3$

7) $w = 1 - 0.1(-6) = 1 + 0.6 = 1.6$

$b = 0 - 0.1(-3) = 0.3$

8) $\hat{y} = wx + b = 0$

$\hat{y} - y = -2$

$\frac{\partial L}{\partial w} = -2 \cdot 1 = -2$

$\frac{\partial L}{\partial b} = -2$

$w_1 = 0 - 0.5(-2) = 1$

$b_1 = 0 - 0.5(-2) = 1$

$\hat{y} = 1 + 1 = 2$

$\hat{y} - y = 0$

$\frac{\partial L}{\partial w} = 0 \quad \frac{\partial L}{\partial b} = 0$

$w_2 = 1 - 0 = 1$

$b_2 = 1 - 0 = 1$